miércoles, 24 de septiembre de 2025 09:46 a. m.

Method I

This is an **extrinsic method**, in the sense that it requires determination of the value of a **specific parameter by the user**.

It involves the definition of a function $h(C)$ that measures the **dissimilarity** between the vectors of the same cluster C. That is, we can view it as a **self-similarity measure**.

For example,

$$\begin{aligned} h_1(C) &= \max\{d(x,y),\ x,y \in C\} \\ h_2(C) &= \text{med}\{d(x,y),\ x,y \in C\} \end{aligned}$$

When d is a metric distance, $h(C)$ may be defined as

$$h_3(C) = \frac{1}{2|C|} \sum_{x \in C} \sum_{y \in C} d(x,y)$$

Let θ be an appropriate **threshold** for the adopted $h(C)$.

Then, the algorithm terminates at the R_t clustering if

$$\exists C_j \in R_{t+1} :\ h(C_j) > \theta$$

In words, R_t is the **final clustering** if there exists a cluster C in R_{t+1} with dissimilarity between its vectors greater than θ.

Sometimes the threshold θ is defined as

$$\theta = \mu + \lambda\sigma$$

where μ is the **average distance** between any two vectors in X and σ is its **variance**.

The parameter λ is a **user-defined parameter**.

Thus, the need for specifying an appropriate value for θ is transferred to the choice of λ. However, λ may be estimated more reasonably than θ.

Lifetime of a cluster

An intuitive approach is to search in the proximity dendrogram for clusters that have a **lifetime**.

The lifetime of a cluster is defined as the absolute value of the difference between the **proximity level** at which it is created and the proximity level at which it is absorbed into a larger cluster.

However, **human subjectivity** is required to reach conclusions.

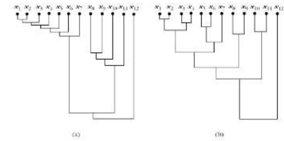

Figure: Example

Method II

This is an **intrinsic method**; that is, in this case only the structure of the data set X is taken into account.

According to this method, the final clustering R_t must satisfy the following relation:

$$d^{ss}_{\min}(C_i, C_j) > \max\{h(C_i), h(C_j)\}, \qquad \forall C_i, C_j \in R_t$$

In words, in the final clustering, the **dissimilarity between every pair of clusters** is larger than the **self-similarity** of each of them.

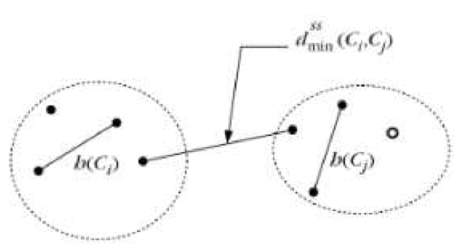

Figure: of the termination condition for method II